# sinfonion

## _travelers guide
## _for piloting your starship

**5** sections · **8** pitch outs · **98** voicings

**[ inv ]** press  **[ inv ]** hold  › then  noon = 12 o'clock  cw / ccw = clockwise / counter

harmonic cv processor · three quantizers, chords, arpeggiator and a chord progression sequencer · one key for all



## 01 five sections

sf_01 · one key for all

| | |
|---|---|
| channels 1–3 | quantizers: IN › nearest allowed note › OUT |
| chords | four voices on OUT 1–4 |
| arpeggio | one voice, a new note on every CLK |

## 03 the knobs

sf_03

| | |
|---|---|
| **[ root ]** **[ degree ]** | the key · the chord / scale in the mode |
| **[ trans ]** | ±12 semitones on every output |
| **[ slew ]** **[ preset ]** | glide time (ch 1 + chords buttons) · 16 presets, or the song with SEQ |

## 02 note buttons

sf_02 · fig.01

| | |
|---|---|
| **[ root ]** **[ 3rd ]** **[ 5th ]** **[ 7th ]** | scale notes 1 · 3 · 5 · 7 |
| **[ 9th ]** **[ 11th ]** **[ 13th ]** | scale notes 2 · 4 · 6 |
| **[ fill ]** | the notes outside the scale · ch 2 has FILL 1–5 · all lit = chromatic |

## 07 modes

sf_07 · **[ mode ]**

| | |
|---|---|
| chords | lyd · maj · 7 · sus · alt · hm5 · dor · min · hm · phr · dim · aug |
| chords2 | 7#11 · 7♭13 · sus4♭9 · 7#9 · -△7 · △7#5#11 · -7♭5 |
| slashch. | a triad over a bass note · the bass stays lowest |
| cadences | major · minor · mel. / har. minor · har. / dbl. har. major: DEGREE walks the key |

## 05 chords

sf_05 · out 1–4

| | |
|---|---|
| voicing | first four lit buttons › OUT 1–4 · fewer = doubled |
| **[ pitch ]** **[ spread ]** | lowest note · how far apart the four sit |
| **[ inv ]** | off = nearest the input · else steps the inversions |
| **[ inv ]** + note | ROOT / 5TH / 9TH / 13TH = 3rd / 2nd / 1st / root position |

## 08 sequencer

sf_08 · fig.02

| | |
|---|---|
| **[ seq ]** | on: PRESET picks the song · **[ ▶/❚❚ ]** play / pause |
| a step | a chord, its beats, all 52 buttons + octaves |
| **[ add ]** **[ del ]** | copy · delete · **[ ← ]** **[ → ]** move · **[ − ]** **[ + ]** beats |
| **[ clk ]** | a beat per trigger, ÷1–64 per part |
| 2× tap **[ ▶/❚❚ ]** | **freeze**: set up the next chord silently |

## 10 oct page

sf_10 · **[ oct ]**

| | |
|---|---|
| octave | ROOT up · 5TH down, ±3, in every section |
| bypass | ch 1–3 13TH · dim = through + transpose · off = true |
| in follows root | FILL · FILL 1 · EQ-DIST · INV · PATTERN |
| 1/2/3/5 V oct | SLEW · FILL 5 · 11TH · chords SLEW · BEGIN |
| glide detect | ch 1 B ROOT · ch 2 FILL 2 · ch 3 3RD |
| **[ gate 2 ]** | **matrix page**: top 4×4 mixes INs into ch 1–3 + arp, next 4×4 routes their OUTs · **[ gate 1 ]** back |

## __fig.01 : scale = chord

why 9TH means the 2nd

C major as a seven-note chord: stack every second note

| C | D | E | F | G | A | B | C | D | E | F | G | A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ROOT | | 3RD | | 5TH | | 7TH | | 9TH | | 11TH | | 13TH |

so 9TH = 2nd note · 11TH = 4th · 13TH = 6th · each button means a scale step, not a fixed pitch

## __fig.02 : a song

**[ seq ]** **[ song ]**

**16 songs** (PRESET knob) → **4 parts** (A · B · C · D) → **32 steps** (one chord each) → **1–128 beats** (a CLK each, + menu)

display ○●○○ beats  ■□■□□□ steps · filled = now

each step remembers its chord, length, all 52 buttons and the octave switches · edits save at once

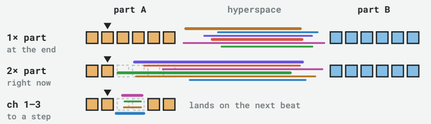
song page

ch 1–3 buttons = steps 1–32

## 09 song page

sf_09 · **[ song ]** · jump to hyperspace

| | |
|---|---|
| ch 1–3 | the 32 steps · press = jump, on the beat |
| chords | ROOT 3RD 5TH 7TH = parts A–D · 1× next · 2× now |
| arp sync | **live mode**: buttons stop saving to steps |




## 11 cv 1–4 · pots

sf_11 · menu

| | |
|---|---|
| global | transpose · note shuffle · detune · harmonic shift (5 V = root only) · preset · slew · voltmeter |
| sections | pre-transpose · pitch bend · spread · voicing · arp clock / random |
| sequencer | song · part (1 V each) · song + part · step |
| **[ pot 1 ]** **[ pot 2 ]** | any cv function · on a cv's function = its attenuator |

## 12 in · out 1/2

sf_12 · menu

| | |
|---|---|
| **[ in 1/2 ]** | slew · quantize now · chords shuffle · arp reset · song / part reset · step / part ± · sync slave |
| **[ out 1/2 ]** | autogates ch 1–3 · arp beat · seq clock / beat / step / reset · tuning · ch 2 key gate · sync master |

## 04 quantizers

sf_04 · ch 1–3

| | |
|---|---|
| **[ trig ]** | patched = sample & hold on each rise |
| ch 1 a / b | B on odd beats, clocks or the A/B jack, e.g. root on 1 + 3 |
| ch 2 | FILL 1–5 = all 12 notes · single note mode = a keyboard |
| **[ eq-dist ]** | ch 3: each lit note gets an equal share of the input |

## 06 arpeggiator

sf_06 · fig.04

| | |
|---|---|
| **[ clk ]** | a note per trigger · fed from the seq CLK |
| **[ pitch ]** **[ range ]** | lowest note · span, up to 5 octaves |
| **[ pattern ]** | 8 patterns (fig.04) |
| **[ begin ]** | restart from that note on each chord change |
| arp clock | ÷4 to ×4, 3/2 = triplets (menu) |

## 13 tune · intonation

sf_13

| | |
|---|---|
| **[ tune ]** | every pitch out plays a C · **[ ← ]** **[ → ]** −2 to +8 V, about 1 mV true |
| intonation | equal · just, retuned to each root · quarter-comma meantone |

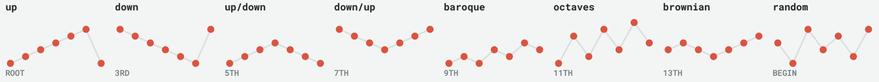
## 14 sync · firmware

sf_14

| | |
|---|---|
| link | master OUT 1 "sync master" › slave IN 1 "sync slave" · any number |
| slaves | follow key, scale, trans + clock · own knobs off · "Sync: dead" if it drops |
| update | sd dongle, aclsinf1.fw · blue ×3 = done |
| outs | 8 pitch, −2 to +8 V, 16-bit · 4 gates |

## __fig.04 : eight patterns

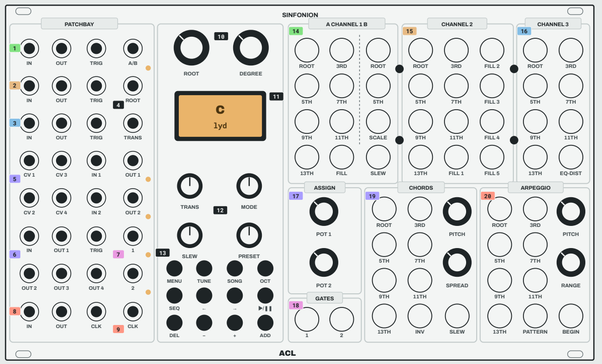
**[ pattern ]**




hold PATTERN + the button under a pattern to jump to it · octaves may pass RANGE · random never repeats a note

## __fig.03 : front panel

20 areas · tags in section colours




| | | | |
|---|---|---|---|
| 1 | **ch 1 jacks** · in · out · trig · a/b | 11 | **display** · chord · scale · beats |
| 2 | **ch 2 jacks** · in · out · trig | 12 | **trans** · **mode** · slew · preset |
| 3 | **ch 3 jacks** · in · out · trig | 13 | **transport** · menu · tune · song · oct · seq … |
| 4 | **root** · **trans** · key + transpose cv | 14 | **channel 1** · a + b sides |
| 5 | **assign** · cv 1–4 · in · out 1/2 | 15 | **channel 2** · all 12 notes |
| 6 | **chords jacks** · in · trig · out 1–4 | 16 | **channel 3** · eq-dist |
| 7 | **gate outs** · gates 1 + 2 | 17 | **pot 1** · **2** · assignable |
| 8 | **arp jacks** · in · out · clk | 18 | **gates** · gate 1 + 2 on / off |
| 9 | **seq clk** · beats · normalled to arp | 19 | **chords** · voicing · inv · slew |
| 10 | **root** · **degree** · the key + chord | 20 | **arpeggio** · pattern · begin |

## _where things hide

almost nothing · sf_15

### hold shortcuts

- **pattern + note** pick a pattern
- **begin + note** pick the start note
- **inv + note** pick an inversion
- **2× ▶/❚❚** freeze

### guarded on purpose

- **tune + ← + ch 1 3rd** calibration · MENU leaves
- **ch 3 3rd 7th 11th eq-dist + arp begin** factory reset, no undo
- **menu › reset to defaults** + twice · keeps songs

### button lights

| | |
|---|---|
| **note buttons** | ● sounding ● selected ○ off |
| **chords** | ● past the first four |
| **oct page** | ●● root / 5th = up ●● centre ●● down |
| **song page** | ● a step ● now blink part queued |